

Marllon Rodrigues

Automation & Control Systems Engineer

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Technical Skills

Industrial Automation: AVEVA Plant SCADA (Citect), System Platform, InTouch, Historian, Power Operation (EPO), MES, FactoryTalk, ISA-101, SEL Relays (F650, 869, 845), Danfoss VFDs, Schneider M580 PLCs, Siemens S7-1200/1500, Rockwell (Control/Compact/MicroLogix), IEC 61850 MMS, Modbus TCP, Profinet, Ethernet/IP, OPC, FPSO Systems, IT/OT Integration.
Software & Tools: Python, TypeScript, Node.js, PostgreSQL, Docker, CI/CD (GitHub Actions), C# (.NET), Git, Prisma.

Experience

Automation Analyst, Tyconnex

Feb 2026 – Present

- **FPSO Commissioning (Petrobras P80/P83):** On-site commissioning engineer for AVEVA Plant SCADA (Citect) on two identical offshore platforms — 50,000+ tags, 1,700+ field devices, Schneider M580 PLCs, 168 panels across 5 industrial subnets (MV 13.8kV, 6.6kV, LV, AUX, MULTIPURPOSE).
- **VFD Integration & Troubleshooting:** Commissioned and validated Danfoss VFDs (02A/02B/03A/03B) via Modbus TCP; diagnosed W1007/W1022 faults through live register-level analysis. Commissioned BT19 Ruhrpumpen MV 6.6kV block (SCADA_READ validated, write control complete).
- **TGCP & IEC 61850:** Commissioned TG-5147001 A/B/C/D gas compressors via Modbus TCP over RTDS; resolved critical tag scaling fault (Eng Full = 10,000,000, divide-by-100 factor on TG02). Configured and validated SEL protection relays (F650, 869, 845) via IEC 61850 MMS — 7,000+ signals catalogued across 105 IEDs.
- **ESA Network Audit:** Physical survey of 77 switch actions validated; identified and reconciled 68 discrepancies across Hirschmann switches (TES001–TES027) including rogue FMM devices flagged as P0.
- **Field Engineering Tool:** Built a custom offline TagFinder app (single HTML, 4.58MB) — 3,432 P80 + 1,785 P83 equipment records across 168 panels, Modbus address tables, IEC 61850 signal catalog, IP mapping for 1,339 devices; used daily by field team for real-time troubleshooting.
- **RDO Generator:** Automated Vendor Daily Report pipeline (Python + PyYAML + LaTeX + Docker); 27+ professional PDF reports generated, documenting 231 hours of on-site commissioning work.

Full Stack Engineer, Bunkerchain (Singapore/Remote)

Jun 2025 – Present

- **Maritime Platform (MVP 3.0):** Full-stack development on Marine Vessel Pass — a Port Identity Layer for international shipping covering vessel onboarding, port calls, bunkering, and digital compliance; FSM-driven port call lifecycle with regulatory holds and multi-signatory workflows.
- **VPS Cryptographic Seal Verification:** Designed and built a trust verification system for physical fuel sample bottle seals — ECDSA digital signatures, public key registry, seal lifecycle state machine, 6 REST API endpoints.
- **RBAC Security Hardening:** Closed 10+ cross-tenant authorization bypass vulnerabilities across 12+ user roles; enforced step-up authentication on all sensitive operations (sign, seal, revoke); implemented hash-chained audit logs with global sequence integrity.
- **Surveyor Identity System:** Built Licensed Surveyor role with profile, device management, and roster UI; 186+ automated tests for comprehensive RBAC coverage.
- **eBDN Platform (Methanol, delivered):** End-to-end delivery of the Terminal Role feature on the electronic Bunker Delivery Note system — from architecture to live client demo approval.

Systems Developer, MAIMA Technology

Aug 2024 – May 2025

- **SAP Automation (RPA):** Development of Python robots for automation of 4–6 critical daily SAP processes, reducing manual execution time from 8 hours to 2 minutes.
- **Predictive Maintenance:** C# system for machine downtime prediction based on product quality analysis, anticipating critical failures before occurrence.
- **Advanced Architecture:** Distributed microservices (Image Analysis, Data, BFF); AI-powered CI/CD pipelines with automated code review and security scanning on every commit; Z-score and ETS forecasting for automatic anomaly detection.

Automation Technician, S.Build Automation

Jun 2022 – Aug 2024

- **Field Engineering & Commissioning:** Specialist in implementing AVEVA System Platform ecosystems, InTouch interface development, and Historian configuration in critical industrial environments — extensive on-site commissioning travel.
- **Industrial Tech Stack:** Projects with Siemens PLCs (S7-1200/1500), Rockwell (Control/Compact/MicroLogix), and

Schneider. Networks: Profinet, Ethernet/IP, Modbus, OPC. Workflows: **AVEVA WorkTasks**.

- **Data Automation & Retrofit:** Python scripts for mass SQL updates in AVEVA MES; legacy migration (1995 → 2023) and reverse-engineering of undocumented machines; upgrade of 20+ servers via ThinManager.

WorldSkills Competitor (Mechatronics), SENAI Roberto Simonsen

Sep 2021 – Feb 2023

- **State Champion (São Paulo Skills):** Selected as **Top 1 Student** of SENAI Roberto Simonsen for the elite team; team became **State Champion**. Training: 8–10 daily hours of precision assembly, PLC programming (Siemens S7-1500), Festo MPS stations, and Industry 4.0 integration via IO-Link.

Certifications

- **CSI AVEVA** Manufacturing Execution System (MES) 2020 – Operations & Performance Exams
- **CSI AVEVA** Application Server 2023 Exam
- **.NET Developer Bootcamp** (DIO – 93h) **Java OOP** (Udemy – 54.5h) **Node.js Fundamentals** (DIO – 30h)

Education

UNINTER, Systems Analysis and Development	2023 – 2025
SENAI Roberto Simonsen, Computer Networks Technician	2021 – 2022
SENAI Roberto Simonsen, Electronics Maintenance Electrician	2020 – 2022

Languages

- English (C1 – Advanced)
- Portuguese (Native)